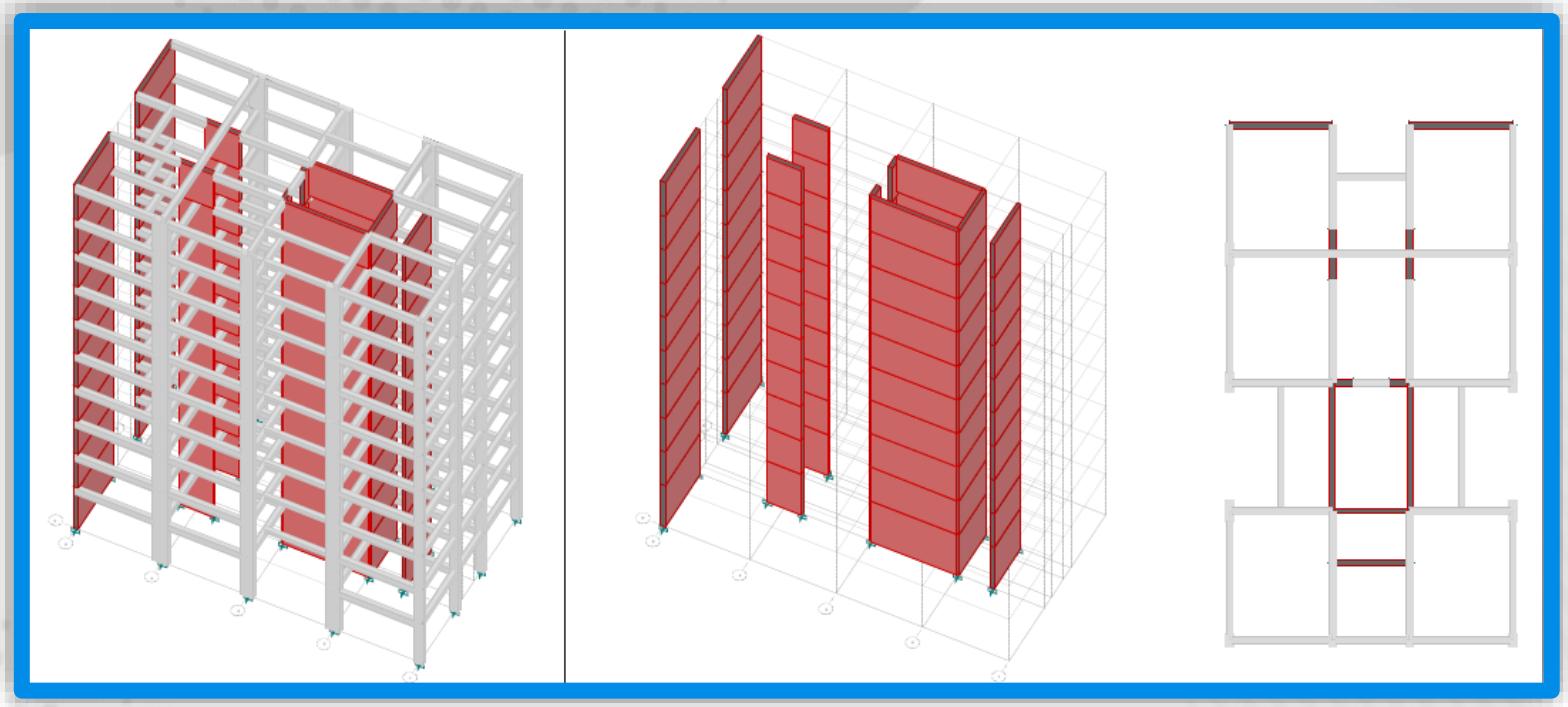


ANÁLISIS Y DISEÑO DE EDIFICACIONES DE CONCRETO ARMADO CON ETABS

SESIÓN 3 – FACTORES DE REDUCCIÓN DE RESISTENCIA

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FACTORES DE REDUCCIÓN DE RESISTENCIA

Table 21.2.1—Strength reduction factors ϕ

Action or structural element		ϕ	Exceptions
(a)	Moment, axial force, or combined moment and axial force	0.65 to 0.90 in accordance with 21.2.2	Near ends of pretensioned members where strands are not fully developed, ϕ shall be in accordance with 21.2.3.
(b)	Shear	0.75	Additional requirements are given in 21.2.4 for structures designed to resist earthquake effects.
(c)	Torsion	0.75	—
(d)	Bearing	0.65	—
(e)	Post-tensioned anchorage zones	0.85	—
(f)	Brackets and corbels	0.75	—
(g)	Struts, ties, nodal zones, and bearing areas designed in accordance with strut-and-tie method in Chapter 23	0.75	—
(h)	Components of connections of precast members controlled by yielding of steel elements in tension	0.90	—
(i)	Plain concrete elements	0.60	—
(j)	Anchors in concrete elements	0.45 to 0.75 in accordance with Chapter 17	—

COMMENTARY

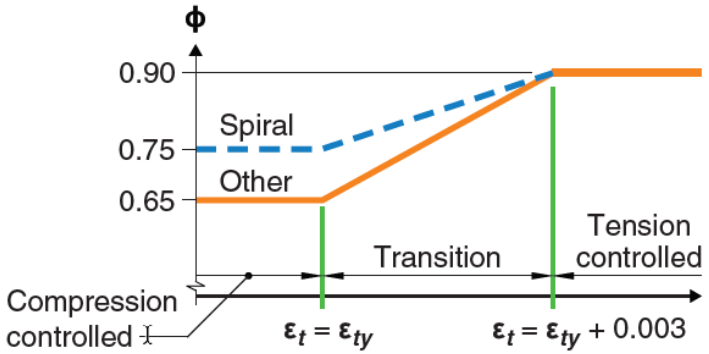
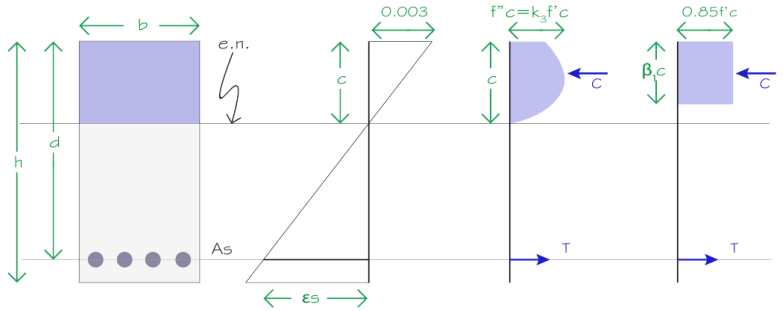


Fig. R21.2.2b—Variation of ϕ with net tensile strain in extreme tension reinforcement, ε_t.

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DISEÑO DE VIGAS



$$M_u \leq \phi M_n \tag{15}$$

$$a = d - \sqrt{d^2 - \frac{2 \cdot M_u}{0.85 \cdot f'_c \cdot \phi \cdot b}} \tag{16}$$

$$c_{max} = \frac{\varepsilon_c}{\varepsilon_c + \varepsilon_{smin}} \tag{17}$$

$$a_{max} = \beta_1 \cdot c_{max} \tag{18}$$

$$\text{if } a \leq a_{max} \Rightarrow \text{Acero sólo en tracción} \tag{19}$$

$$\text{else} \Rightarrow \text{Necesita acero en compresión}$$

$$A_s = \frac{M_u}{\phi \cdot f_y \cdot (d - \frac{a}{2})} \tag{20}$$

$$\rho_{max} = 0.75\rho_b \tag{21}$$

$$A_{smax} = \rho_{max} \cdot b \cdot d \tag{22}$$

$$A_{smin} = 0.80 \cdot \frac{\sqrt{f'_c}}{f_y} \cdot b \cdot d \tag{23}$$

$$A_{smin} = \frac{14}{f_y} \cdot b \cdot d \tag{24}$$

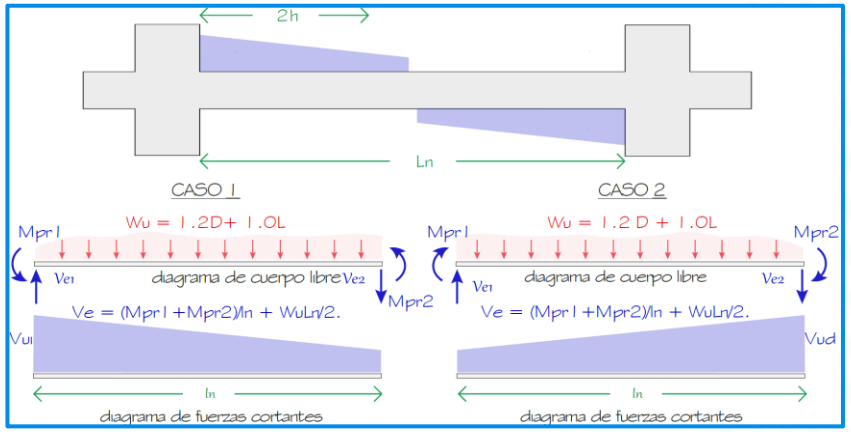
$$\tag{25}$$

$$V_u \leq \phi V_n \tag{26}$$

$$V_n = V_c + V_s \tag{27}$$

$$V_c = 0.53\sqrt{f'_c} \cdot b \cdot d \tag{28}$$

$$V_s = \frac{Av \cdot f_y \cdot d}{s} \tag{29}$$



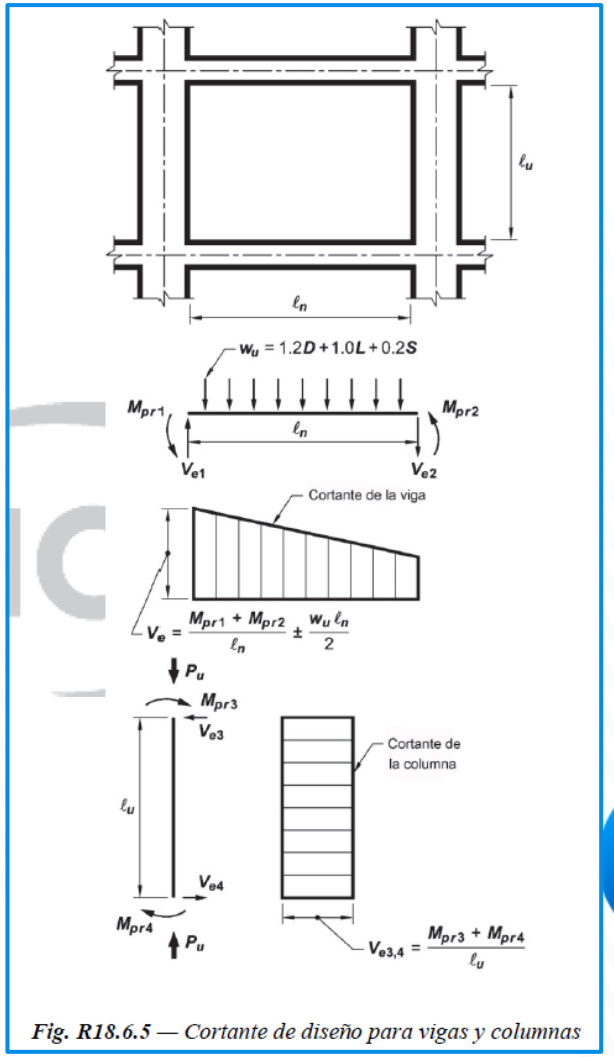


Fig. R18.6.5 — Cortante de diseño para vigas y columnas

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DETALLAMIENTO DE VIGAS

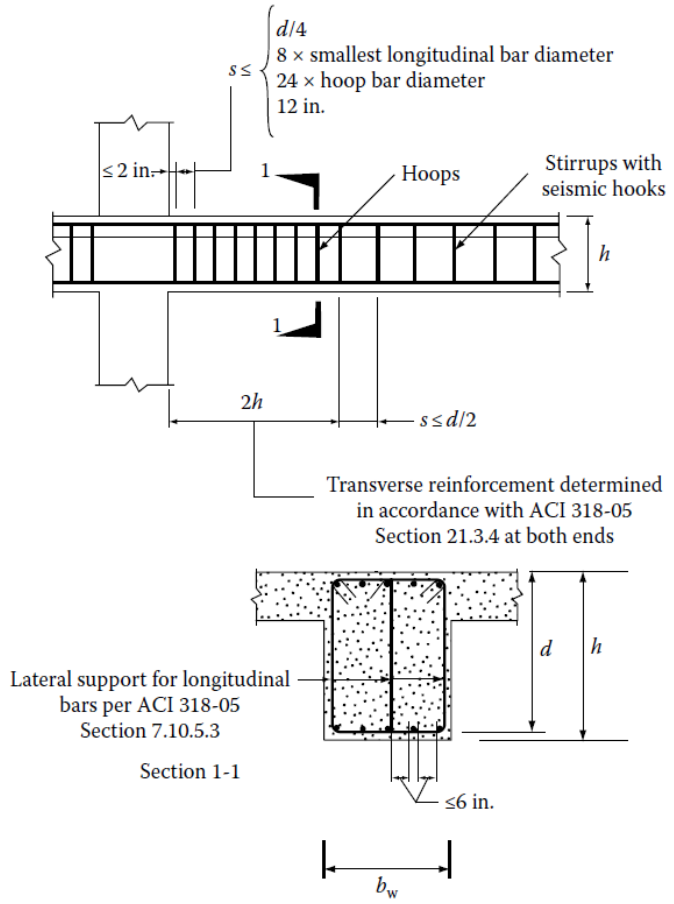


FIGURE 6.13 Frame beam: transverse reinforcement requirements, special moment frame.

REDUCCION POR TORSION

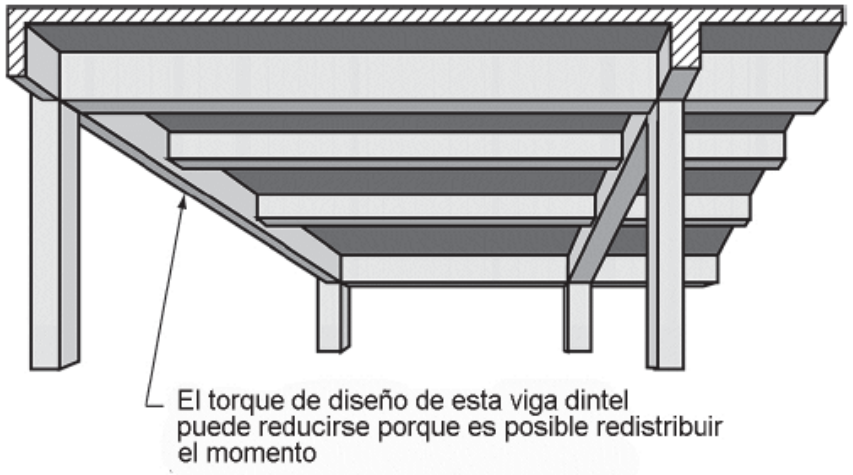


Fig. R22.7.3(b) — *El torque de diseño puede ser reducido (22.7.3.2).*

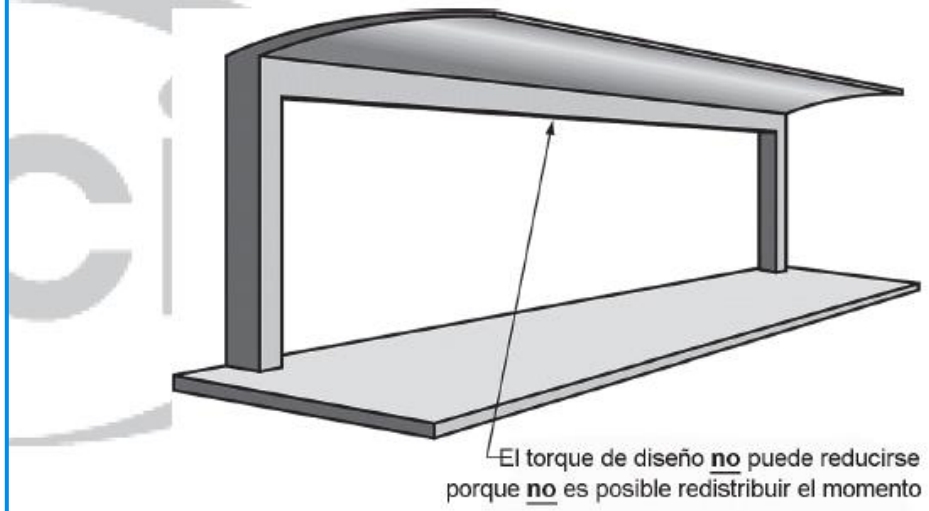


Fig. R22.7.3(a) — *El torque de diseño no puede ser reducido (22.7.3.1).*

DETALLAMIENTO DE COLUMNAS

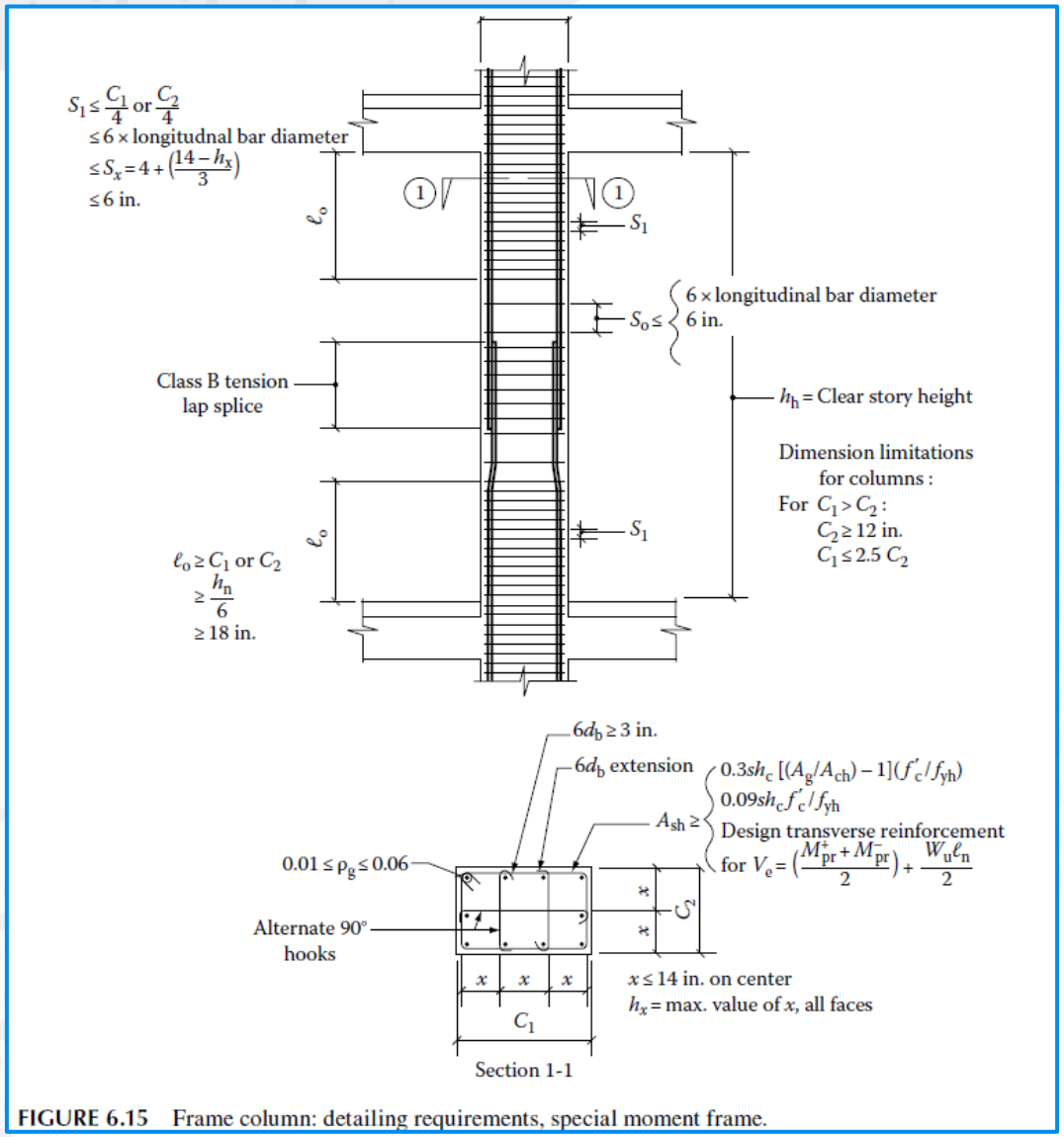


FIGURE 6.15 Frame column: detailing requirements, special moment frame.

DETALLAMIENTO DE MARCO

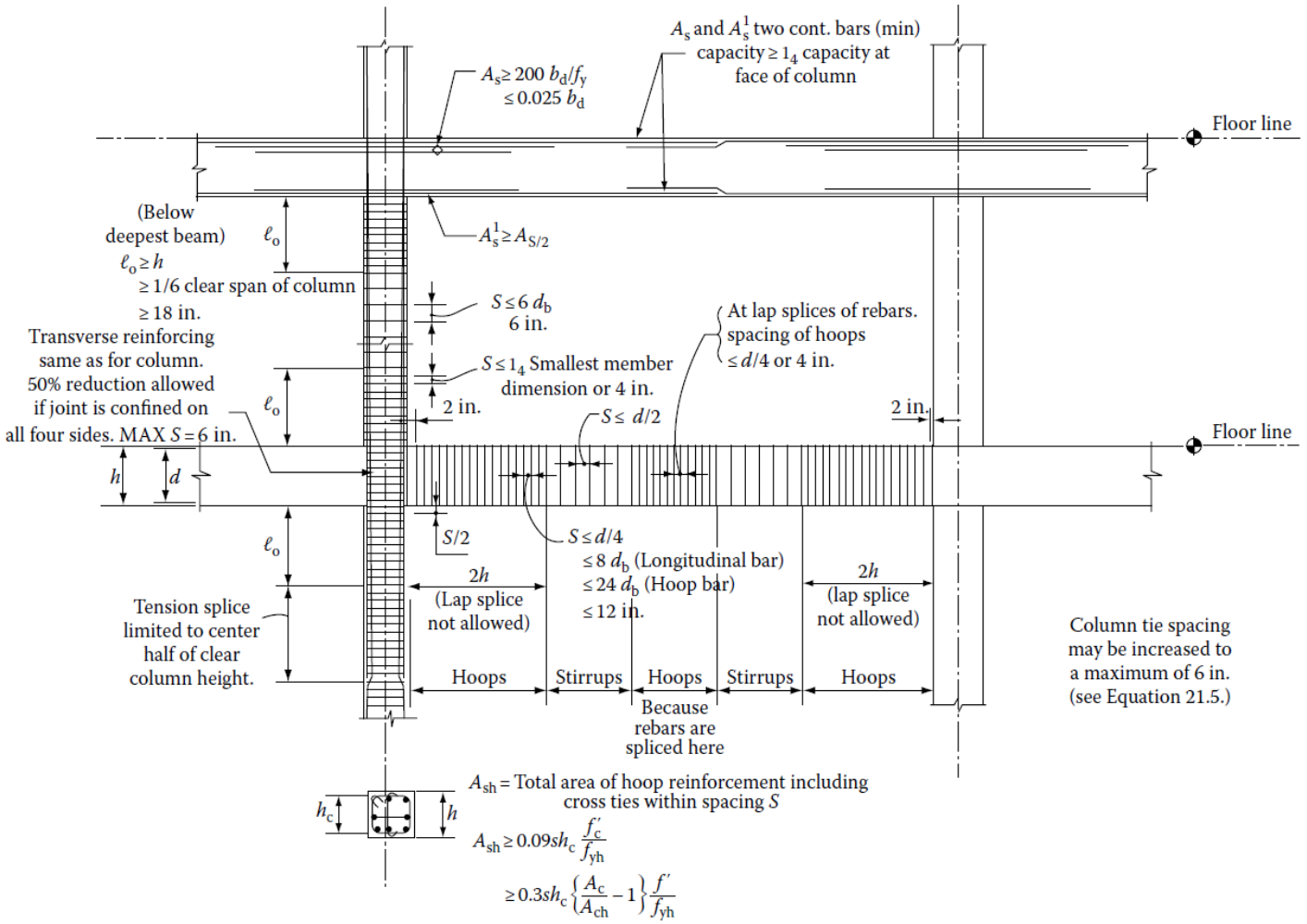


FIGURE 6.17 Ductile frame, SMRF: schematic reinforcement detail.

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